

DATA266 Lab 1

From-Scratch Language Models, Sentiment Classification, and CycleGAN

Team 44 - Parth Patel and Chelsi Shulamite Elthuri

Repository: https://github.com/shulamite512/DATA266_LAB1_Pair44

1. Team Ownership and Repository

Parth Patel implemented, trained, evaluated, and documented the work under the parth/ folders for Tasks 1-3, including the shared reproducibility logs and manifests for his runs. Chelsi Shulamite Elthuri implemented, trained, evaluated, and documented the work under the shulamite/ folders for Tasks 1-3. Each member trained and evaluated their own models; no jointly executed cross-member evaluation exists. The team shared the raw dataset bundle and applied the course-provided Task 3 evaluation protocol independently. Seeds were 8503 (Parth) and 266 (Chelsi).

Table 1. Model ownership.

Member	Task 1	Task 2	Task 3
Parth Patel	Character-level GPT	TF-IDF LR, MeanPoolMLP, TextCNN	CycleGAN EXP-05
Chelsi Shulamite Elthuri	Character-level GPT	Mean-pool baseline, TextCNN, BiGRU	CycleGAN run2 (epoch 95)

The results reported here come from the saved metrics, logs, checkpoints, and notebook outputs for each member. When the two members used different datasets or evaluation settings, we treat the comparison as descriptive rather than as a direct ranking.

2. Task 1 - Character-Level GPT from Scratch

Both members built decoder-only character-level GPT models without prebuilt Transformer or attention modules. The implementations share a 128-character context, learned positional embeddings, pre-LayerNorm blocks, dropout 0.1, and gradient clipping at 1.0, but differ substantially in model size, training data, optimization, and hardware.

2.1 Architecture and training setup

Table 2. Task 1 configurations.

Attribute	Parth	Chelsi
Dataset	TinyStoriesV2-GPT4; 80k loaded	Original TinyStories; 150k story pool
Train / validation	100k / 10k windows; 12.8M / 1.28M chars	100k / 10k stories; 89.1M / 8.9M chars
Vocabulary	81	109
Width / layers / heads	256 / 4 / 8	96 / 3 / 4
FFN	1024 (4×)	192 (2×)
Parameters	3,233,792	257,869
Optimizer	AdamW, lr 3e-4, betas 0.9/0.95	AdamW, lr 3e-5, default betas
Schedule	5% warmup + cosine; 10 epochs	0.25-epoch warmup + cosine; 20 epochs
Batch / precision	128 / bf16	32 / fp32
GPU	RTX 5090	RTX 3080 Ti Laptop
Seed	8503	266

Parth used a much larger model, while Chelsi trained on substantially more characters. Since the two runs used different TinyStories files and different vocabularies, the loss, perplexity, BPC, and accuracy values should not be read as a direct head-to-head comparison.

2.2 Quantitative results

Table 3. Task 1 metrics. CE values are in nats per character.

Metric	Parth	Chelsi	Comparability
Validation CE	0.7394	1.0939	Not controlled: different data/vocab
Perplexity	2.0948	2.9858	Not controlled
BPC	1.0668	1.5781	Not controlled
Top-1 accuracy	0.7655	0.6582	Not controlled
Train CE, eval mode	0.7201	Not reported	-
Generalization gap, eval mode (val – train)	0.0193	Not reported	Canonical gap for Parth
Val – train, dropout-on epoch-mean train CE	−0.0327	−0.0906	Definition aligned; data differ
Greedy rep-4	0.0734	0.7031	Partial: prompts/length differ
Sampled rep-4	0	0	Partial: decoding differs
Max grad norm (pre-clip)	9.55	1.41	Same concept
NaNs / spikes	0 / 0	0 / 0	Stable in both
Training time	91.84 s	11,695 s raw; ~1.97 h adjusted	Hardware/data differ
Peak GPU allocated	1,240 MB	147 MB	Same memory concept

Parth's canonical generalization gap is 0.0193 nats per character, computed with both training and validation cross-entropy in eval mode (train 0.7201, validation 0.7394). The negative values in the dropout-on row compare validation loss against the epoch-mean training loss recorded with dropout active; they are reported only because that definition is available for both members, and they reflect dropout noise rather than validation data being easier.

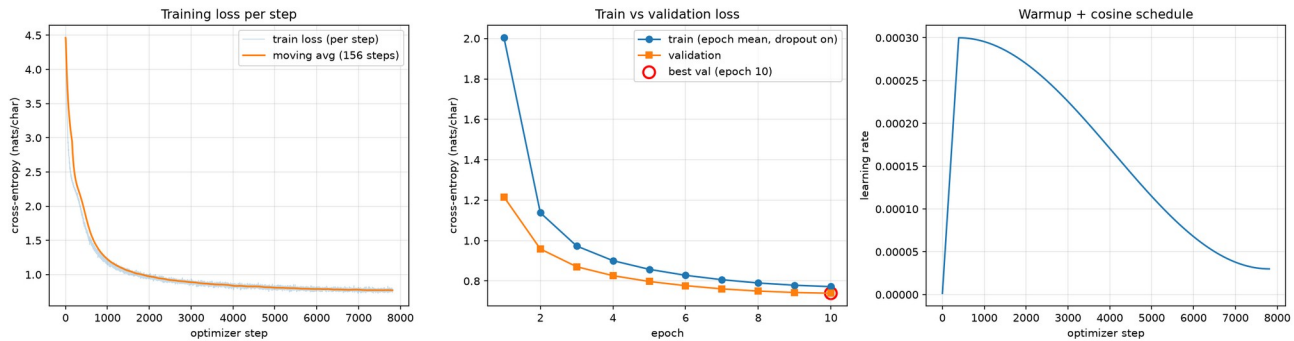


Figure 1. Parth's Task 1 training dynamics: per-step loss, train/validation loss by epoch, and the warmup-cosine learning-rate schedule.

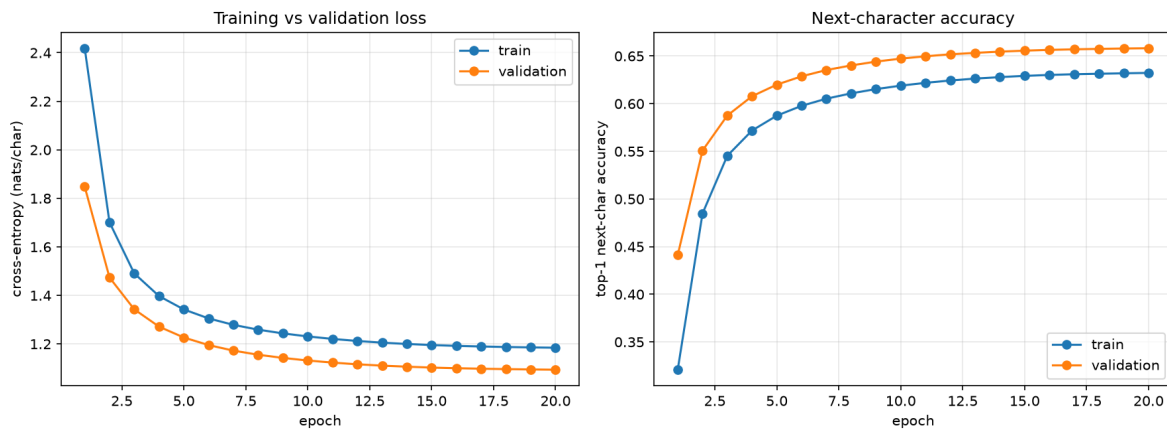


Figure 2. Chelsi’s Task 1 train/validation cross-entropy and next-character accuracy across 20 epochs.

2.3 Generation and failure analysis

Parth generated seven 500-character continuations from his epoch-10 checkpoint: six with temperature 0.8 and top-k 50 (two prompts, three seeds each) and one greedy continuation. The sampled continuations contained no repeated 4-grams ($\text{Distinct-1/2/3} = 0.406/0.812/0.933$), whereas the greedy continuation had a repeated-4-gram rate of 0.073. Table 4 summarizes his three manually reviewed failure cases.

Table 4. Parth’s Task 1 failure cases (committed generated text; [...] marks omitted spans).

Failure type	Short generated snippet	Observation
Entity and name inconsistency (T = 0.8, top-k 50)	“there was a little girl named Mia. [...] Mia was very sad. Lily told Tom to stop and play with him.”	The story switches from Mia to Lily and Tom without introducing them, although Mia is still inside the 128-character window.
Semantic incoherence and broken grammar (T = 0.8, top-k 50)	“Tim had a big purple said,”Hi Tim! You are a nice museum.” [...] his mum would have a raven banana.”	Words are spelled correctly and common in TinyStories but combined without meaning; “a big purple said” drops the noun.
Repetition under greedy decoding	“He wanted to play with it. [...] He wanted to play with it. [...]”I want to play with you, but I want to play with you.””	“to play with it” occurs four times in 500 characters (rep-4 0.073 versus 0.000 for the sampled outputs).

Across Parth’s samples the errors lie in meaning, entity tracking, and repetition rather than spelling, which is consistent with metrics that reward local next-character prediction (top-1 accuracy 0.766, perplexity 2.09).

Chelsi evaluated greedy, temperature-0.8, and temperature-1.0/top-k-20 decoding over 15 samples. Her committed failure analysis (Table 5) identifies greedy repetition loops and noisier, less grammatical text under higher-temperature sampling. In her aggregate metrics, greedy decoding has low diversity ($\text{Distinct-1/2/3} = 0.088/0.118/0.137$) and a repeated-4-gram rate of 0.703, while both sampling settings have zero repeated 4-grams.

Table 5. Chelsi’s Task 1 failure cases (committed generated text).

Failure type	Short generated snippet	Observation
Repetition and broken grammar (greedy)	“and said,”I want to the bird and said, “I want to the bird and said,”I want to the ball ...”	High-probability loop with rep-4 0.836, the highest among her candidates.
Loss of coherence with repetition (greedy)	“he was so happy to the ball and said,”I want to the ball and said, “I want to the bear ...”	A plausible opening is followed by a repeated dialogue template (rep-4 0.765).
Broken grammar and character errors (T = 1.0, top-20)	“So then see burned her wing again. They saw the bull hat a fun. [...] Tim was a limted to his grandma boom.”	Sampling removes the loop (rep-4 0.000) but produces malformed words and abrupt subject changes.

Both models showed the same practical decoding trade-off. Greedy decoding tended to repeat phrases, while sampling improved diversity but sometimes reduced coherence. Validation loss was still falling at the

end of both runs, so additional training may have helped. Reasonable next experiments would be a longer context, repetition-aware decoding, and training windows that stay within story boundaries.

3. Task 2 - Yelp Polarity Sentiment Classification

Each member trained three sentiment models from scratch without pretrained embeddings or language models. Parth evaluated on the full official 38,000-review test set, whereas Chelsi evaluated on a seeded, balanced 10,000-review subset. Cross-member ranking is therefore not controlled; the strongest conclusions are within-member comparisons.

3.1 Preprocessing and models

Table 6. Task 2 data and preprocessing.

Attribute	Parth	Chelsi
Train / val / test	531,965 / 28,000 / 38,000	90,000 / 10,000 / 10,000
Text handling	lowercase; apostrophe deletion; non-alnum cleanup	lowercase; HTML strip; n't → not; letters only
Stopwords	sklearn minus 11 negations	28 custom stopwords
Stemming	Porter	Porter
Vocabulary	50,000	32,094
Max length	256 (1.42% of test truncated)	120 (26.0% of test truncated)

Table 7. Task 2 models.

Member	Model	Architecture	Params
Parth	TF-IDF LR	1-2 gram TF-IDF (200k) → logistic regression	200,001
Parth	MeanPoolMLP	Emb128 → mean pool → MLP256	6,433,538
Parth	TextCNN	Emb128 → Conv k3/4/5 ×128 → max pool	6,597,762
Chelsi	Mean baseline	Emb64 → mean pool → MLP64	2,058,241
Chelsi	TextCNN	Emb96 → Conv k3/4/5 ×96 → max pool	3,192,193
Chelsi	BiGRU	Emb64 → bidirectional GRU64 → linear	2,104,065

3.2 Results and statistical tests

Table 8. Task 2 test metrics. Parth n = 38,000 (official test set); Chelsi n = 10,000 (balanced subset).

Accuracy shown with its 95% CI.

Member / model	Accuracy [95% CI]	Macro-F1	ROC-AUC	PR-AUC	MCC	Brier	ECE
Parth LR	0.9463 [0.9441, 0.9484]	0.9463	0.9876	0.9881	0.8927	0.0424	0.0321
Parth MeanPool	0.9313 [0.9286, 0.9337]	0.9313	0.9803	0.9804	0.8626	0.0517	0.0048
Parth TextCNN	0.9379 [0.9354, 0.9401]	0.9379	0.9848	0.9853	0.8759	0.0461	0.0059
Chelsi Mean	0.9119 [0.9065, 0.9173]	0.9119	0.9705	0.9703	0.8238	0.0646	0.0069
Chelsi CNN	0.9143 [0.9089, 0.9197]	0.9143	0.9755	0.9755	0.8288	0.0621	0.0271
Chelsi BiGRU	0.9258 [0.9206, 0.9312]	0.9258	0.9785	0.9791	0.8520	0.0553	0.0165

Within Parth's experiments, TF-IDF logistic regression was the strongest model on accuracy, macro-F1, ROC-AUC, and MCC. Its advantage over the MeanPoolMLP and the TextCNN was statistically significant under continuity-corrected McNemar tests ($p = 2.39\text{e-}55$ and $p = 2.81\text{e-}18$, respectively). The MeanPoolMLP

had the lowest ECE (0.0048; TextCNN 0.0059). The logistic regression was underconfident-accuracy exceeded mean confidence in all 15 reliability bins (ECE 0.0321)-yet it still had the lowest Brier score (0.0424), so a lower ECE did not imply better probabilistic predictions overall (Figure 4).

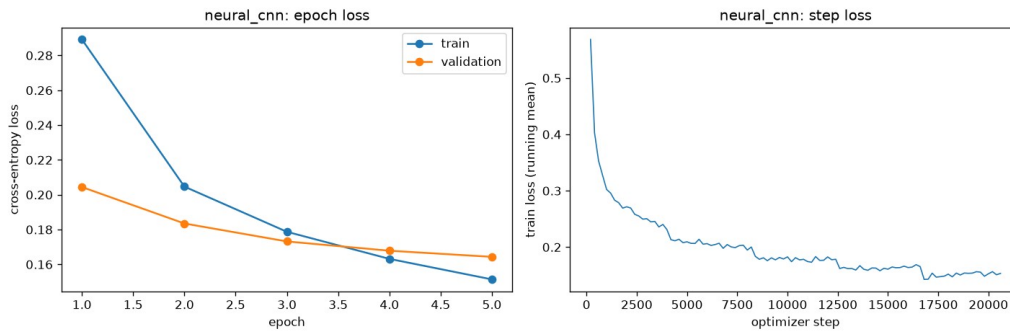


Figure 3. Training and validation loss for Parth's TextCNN.

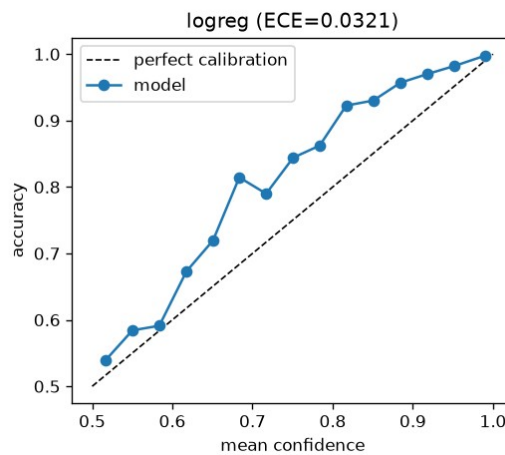


Figure 4. Reliability diagram for Parth's TF-IDF logistic regression (15 bins over [0.5, 1]; ECE 0.0321). Accuracy lies above mean confidence in every bin, i.e., the model is underconfident.

Within Chelsi's experiments, the BiGRU was strongest at 0.9258 accuracy and significantly outperformed the mean-pooling baseline under an exact McNemar test ($p = 4.69\text{e-}08$). The CNN's 0.9143 accuracy was not significantly different from the baseline ($p = 0.393$). Her CNN also showed the clearest overfitting, reaching train loss 0.0239 versus validation loss 0.2660 at epoch 4.

Slice definitions differ between members and are reported separately. For Parth, the contrast-word slice had the highest error rate for all three models. For Chelsi, long reviews and reviews containing negation were the most difficult slices. A shared test set and common slice definitions would be required for a valid cross-member paired comparison.

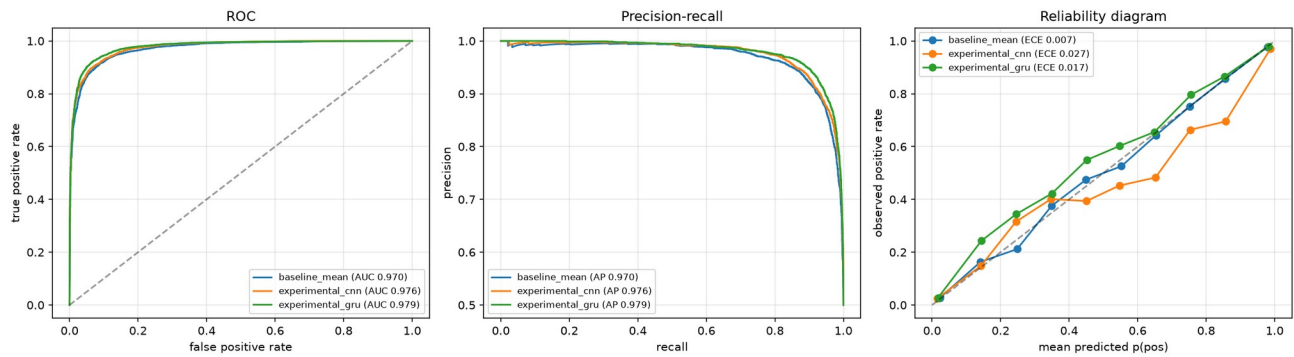


Figure 5. ROC, precision-recall, and calibration views for Chelsi's three Task 2 models.

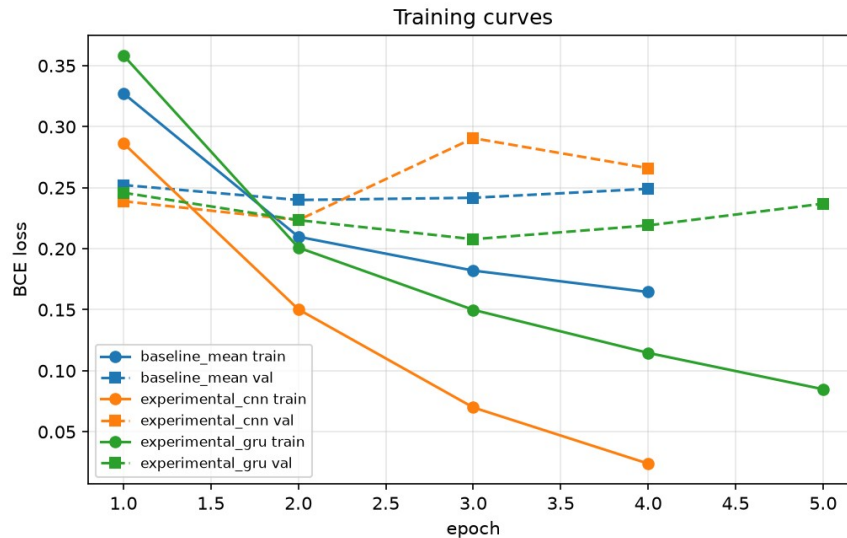


Figure 6. Training and validation BCE for Chelsi's mean-pooling, CNN, and BiGRU models.

3.3 Error analysis and joint discussion

Parth manually reviewed 20 TextCNN errors (seed 8503): five confident false positives ($P(\text{positive}) > 0.998$), five confident false negatives ($P(\text{positive}) < 0.001$), five near-threshold errors ($P(\text{positive})$ between 0.4995 and 0.5006), and five errors from the contrast-word slice, the TextCNN's highest-error slice (error rate 0.0688). Each case was assigned a manual error type and a testable fix. The most frequent category was mixed sentiment or contrast (7 of 20, partly by design because five cases came from the contrast slice), followed by possible rating-text mismatch (3), temporal sentiment shift (2), ambiguous labels on reviews of closing businesses (2), and weak or balanced long reviews (2). Thirteen of the 20 are errors made by all three of his models, and four of those point to labels that the text does not clearly support. A recurring preprocessing side effect is that the stock stopwords list removes contrast words ("but", "however") and number words, so the decisive clause or an explicit star rating is invisible to every model. Table 9 shows one representative case per category.

Table 9. Representative reviewed TextCNN errors (Parth).

Category	Short review snippet	Observed issue	Testable fix
Confident FP (P = 0.9989)	"I have to say I love cafe rio! [...] However I've been to this location 3 different times [...] DO NOT EAT AT THIS LOCATION !"	Mixed sentiment with a target shift; "However" is removed by the stopword list.	Retrain with contrast words kept and compare error rates on the contrast slice.
Confident FN (P = 0.0005)	"All they need to do is add a Forever 21, and then I will never have to go to ghetto Fiesta Mall or drive through god-awful traffic ..."	Sentiment target shift: the negative words describe other places.	Hand-annotate reviews whose negative words refer to other businesses; compare with an order-aware model.
Near-threshold (P = 0.5001)	"Unfortunately it didn't deliver [...] The flavors didn't mix well. [...] I would look elsewhere"	Negation-heavy negative review; a TextCNN-only error (LR 0.327 and MeanPool 0.208 are correct).	Mark negation scope (NOT_ prefix) and compare error on the negation slice.
Contrast slice (P = 0.8485)	"I gave it two stars, but it's because of my tastes, which I will explain in detail."	Balanced review whose explicit rating is lost in preprocessing (reduced to "gave star").	Keep number words, map "N star(s)" to a rating token, and compare error on reviews stating a star count.

Chelsi reviewed 20 BiGRU errors with the same 5/5/5/5 structure and committed a manual error type for each. In her analysis, the confident false positives are mostly reviews with positive local words but a negative overall judgment (sarcasm, faint praise, price/value complaints, or very short context), while the confident false negatives express positive opinions through narrative or mix them with criticism. She describes the near-threshold cases as genuinely ambiguous because different aspects of a business receive different sentiment, and attributes the long-review errors to truncation and dispersed evidence, consistent with her slice results (macro-F1 0.922 without negation, 0.914 on long reviews, and 0.916 on reviews with negation). Her proposed fix preserves negation and discourse structure, increases the sequence length or uses hierarchical pooling, and adds hard negatives, to be tested by retraining with the same seed and split and comparing the long-review and negation slices.

Parth's TextCNN scored higher than his MeanPoolMLP, but we did not run a paired significance test between those two neural models, so that difference is descriptive. Chelsi's BiGRU, in contrast, significantly outperformed her mean-pooling baseline. The error reviews from both members repeatedly point to contrast, negation, and mixed sentiment as difficult cases. A cleaner follow-up would use the same test set for both members, preserve contrast words during preprocessing, reduce truncation, and run direct paired tests between the neural models.

4. Task 3 - CycleGAN Monet/Photo Translation

Both members implemented the same core CycleGAN recipe: ResNet-9 generators, 70×70 PatchGAN discriminators, LSGAN adversarial loss, cycle-consistency weight 10, identity weight 5, replay buffers of 50, and Adam at 2e-4 with $\beta_1 = 0.5$. Their parameter counts are identical (28,285,832), so the main experimental differences are the training schedule, batch size, precision, initialization, seed, and checkpoint selection.

4.1 Architecture and training differences

Table 10. Task 3 configurations.

Attribute	Parth EXP-05	Chelsi run2
Generator / discriminator	ResNet-9 / 70×70 PatchGAN	Same layout
Initialization	N(0, 0.02)	PyTorch default
Schedule	30 constant + 30 decay	50 constant + 50 decay
Epochs / steps	60 / 60,000	100 / 88,000
Batch	1	8
Precision	FP32 (+TF32)	bf16 autocast
Pairs seen	60,000	703,800
Checkpoint	epoch 60	epoch 95
GPU	RTX 5090	RTX 4090
Training time	4,245 s	20,727 s
Peak GPU memory	9,029 MB	9,011 MB

Parth ran a six-experiment campaign around the baseline. None of the single changes in EXP-01 through EXP-04 improved the course combined score over the baseline, and EXP-05 repeated the baseline configuration on a different GPU. Chelsi's final run extended training to 100 epochs, used batch 8 and bf16, and selected epoch 95 using the class-notebook FID sweep. Her run1-to-run2 change bundle improved the course-protocol submission FID from 104.14 to 96.01, but the effects of the changed schedule, augmentation, precision, and selection strategy cannot be separated.

4.2 Course-protocol evaluation

Table 11. Course-protocol metrics (class evaluation notebook).

Metric	Parth EXP-05	Chelsi run2	Interpretation
FID Monet → photo	103.0926	95.8651	Closest shared protocol
FID photo → Monet	103.0202	96.1503	Closest shared protocol
Submission FID	103.0564	96.0077	Chelsi lower by 7.05
Submission MiFID	0.418674	0.408467	Same class-protocol concept
Combined (FID + MiFID)/2	51.7375	48.2081	Lower is better

This is the closest cross-member quantitative comparison, but it still requires caution. Parth ran the instructor evaluator directly, while Chelsi used a reimplementation matching the class notebook. Each model has a single seed, and both checkpoint-selection procedures used the same evaluation set that is reported as the final score. Parth's same-configuration replicate differed by 1.67 FID points, although the replicate also used a different GPU/CUDA environment. The difference therefore cannot be attributed solely to stochastic run-to-run variation. The observed cross-member 7.05-point FID difference is larger than that single replicate difference, but a single, differently executed replicate does not establish the variance of the training procedure.

Internal metrics are not ranked across members because the evaluators, subsets, and metric definitions differ. Parth's internal evaluation reported FID 81.07/85.09, KID 0.0176/0.0126, and density/coverage in both directions (Appendix A). Chelsi's internal evaluation used torchmetrics with first-100 or 300-image subsets for several metrics, producing FID 76.42/77.51 and KID 0.0163/0.0102. These values are informative within each pipeline but not directly comparable.

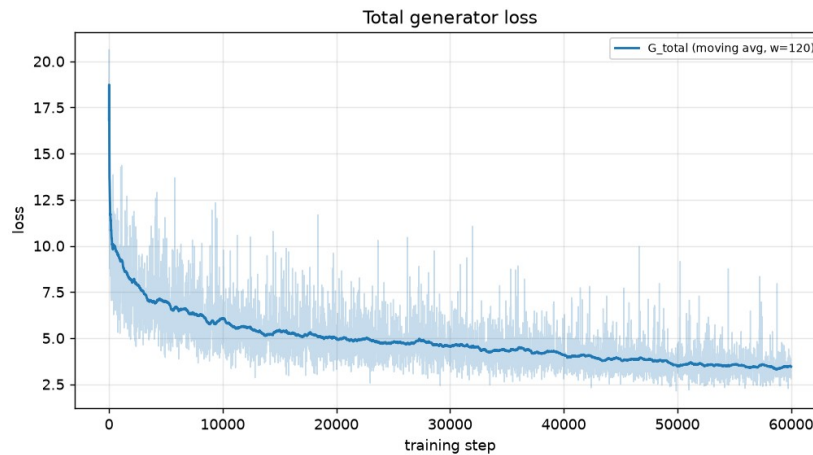


Figure 7. Total generator loss across Parth's 60,000-step EXP-05 run (light: per step; dark: moving average). The loss continues to fall after the linear learning-rate decay begins at epoch 31.

4.3 Qualitative findings, stability, and limitations

Parth's EXP-05 training was numerically stable: no non-finite steps in 60,000, 38 isolated loss-spike steps (loss above three times its moving average), and 6 discriminator-saturated steps, with the largest generator gradient norm (283.95) occurring in epoch 1. The total generator loss fell from 9.27 in epoch 1 to 3.37 in epoch 60 and continued to fall after the learning-rate decay began (Figure 7), while the epoch-60 discriminator losses (D_A 0.142, D_B 0.169) stayed well away from zero. A falling GAN loss shows only the absence of an obvious collapse signature, not image quality. The recorded numbers are consistent with cycle consistency: the epoch-60 training cycle L1 is 0.0707 (A) and 0.0933 (B), the reconstruction L1 over all evaluation images is 0.0807 (Monet → photo) and 0.1018 (photo → Monet), and LPIPS from input to reconstruction (0.2669/0.2307) is lower than from input to translation (0.2987/0.3801). A low cycle error alone does not guarantee a faithful translation.

In the epoch-60 grid (Figure 8), photo → Monet keeps the scene layout while shifting the palette and adding brush-like texture, and the reconstructions are close to the inputs in every row. Parth's failure analysis documents four shortcomings. First, a periodic checkerboard-like texture remains in flat regions such as the sky of the row-7 sunset, a pattern associated with transposed-convolution upsampling; the resize-convolution alternative (EXP-01) reduced it but scored worse overall (combined 57.29). Second, Monet → photo realism is weak: outputs mainly change colour and contrast and retain painterly texture, consistent with that direction's smaller change from the input (LPIPS 0.2987 vs. 0.3801), higher content cosine (0.8147 vs. 0.7525), and low coverage of the real-photo distribution (recall 0.3558, coverage 0.0604). Third, one unusual sunset photo is rendered with an inverted yellow/blue palette; as one image of eight, this documents a failure mode rather than a rate. Fourth, photo → Monet outputs cover the real Monet features well (coverage 0.9467) but many fall outside their neighbourhoods (precision 0.3605). No evidence of mode collapse was observed under the recorded checks: photo → Monet coverage is 0.9467 and recall 0.5367, there were no byte copies or near-blank outputs, and only 1 of 7,038 outputs fell in the near-copy warning band.

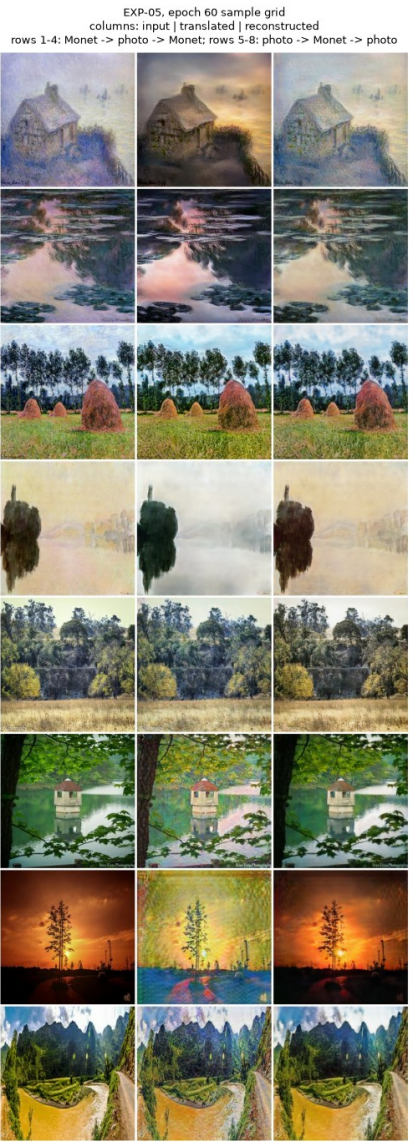


Figure 8. Parth's EXP-05 sample grid at epoch 60. Columns: input, translated, reconstructed. Rows 1-4: Monet → photo → Monet; rows 5-8: photo → Monet → photo. Row 7 is the documented failure case (inverted palette and periodic grid texture in the sky).

Chelsi's run2 completed without NaNs. A short disturbance around epochs 49-51 settled after learning-rate decay began (Figure 9), and the checkpoint sweep in Figure 10 supports the epoch-95 selection. Her final Task 3 write-up and failure analysis are included in the repository, and Figure 11 shows representative translations from the same run.

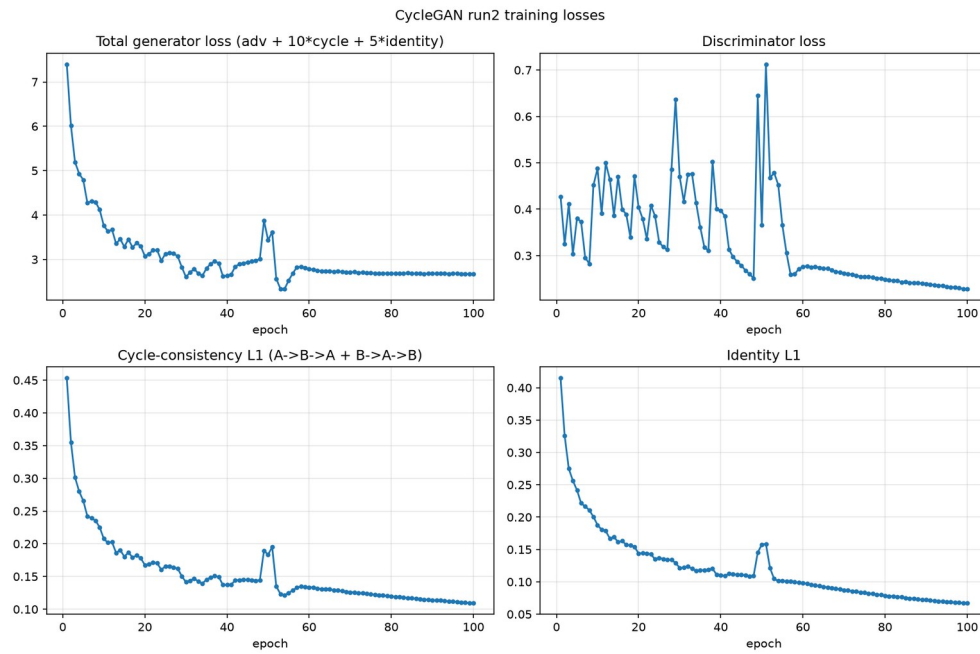


Figure 9. Generator, discriminator, cycle, and identity losses across Chelsi's run2; the epoch 49-51 disturbance is visible.

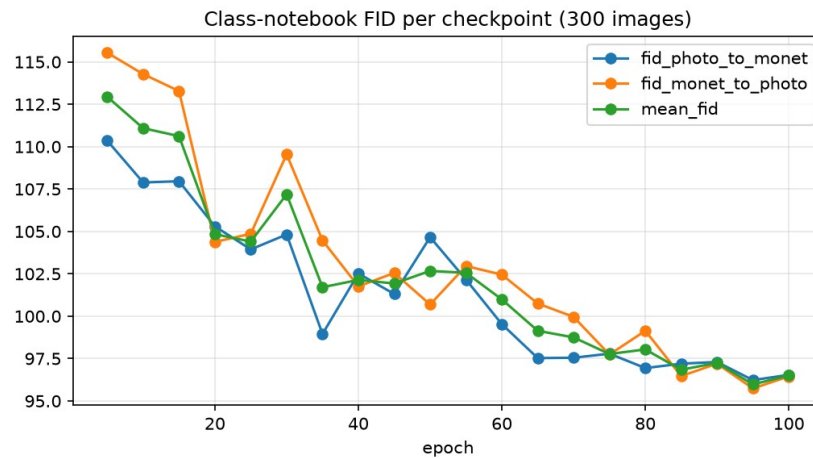


Figure 10. Course-protocol FID by checkpoint for Chelsi's run2, supporting the epoch-95 selection.



Figure 11. Chelsi's run2 translations at the epoch-95 checkpoint. Columns: input photo, photo → Monet, input Monet, Monet → photo; rows: fixed audit samples 1-4.

The main limitations are the single seed used for each final run, checkpoint selection on the same evaluation set used for reporting, and the relatively small 300-image course metric. A stronger follow-up would use multiple seeds, a separate checkpoint-selection set, and controlled ablations for longer schedules and alternative upsampling methods.

4.4 Kaggle submission

The team submitted Chelsi's run2 predictions to the Kaggle class competition as its final entry, because run2 had the lower combined objective of the two members' final models. Kaggle displays the competition objective with a negative sign, so the displayed score of -48.2080 corresponds to the locally computed combined objective of 48.2081 . Parth's EXP-05 (FID 103.0564, MiFID 0.41867, combined 51.7375) was evaluated with the same course protocol but was not the team's final Kaggle entry.

Table 12. Kaggle submission.

Item	Value
Submitted file	submission.csv from Chelsi's run2 (CycleGAN, 100 epochs, epoch-95 checkpoint)
Submission status	Complete
Course-protocol FID / MiFID	96.0077 / 0.40847
Local combined objective (FID + MiFID)/2	48.2081
Kaggle displayed public score	-48.2080
Private score	Not recorded
Leaderboard rank	17 (team PairProgramming_Team_44), as recorded in the repository by the submitting member; no leaderboard screenshot is committed

4.5 Human evaluation

Human evaluation was used alongside the quantitative metrics. During Parth's experiments, generated images were checked against their source images throughout training; for EXP-05, more than 30 pre- and post-training examples were visually reviewed. The outputs generally preserved scene structure while changing palette and texture, especially in the photo-to-Monet direction, while the failure analysis captured checkerboard texture, weaker Monet-to-photo realism, and occasional palette inversion. Chelsi completed the formal 30-sample, two-rater audit for run2 and recorded the style, content, artifact, and inter-rater agreement results in the repository. Parth's development-time review was not a blinded two-rater audit, so no agreement statistic is claimed for EXP-05.

5. Reproducibility and Hardware

Table 13 summarizes the hardware, software, seeds, and checkpoints used for the reported runs. The repository also contains the corresponding configurations, logs, manifests, and metric files.

Table 13. Execution environments and canonical checkpoints.

Member	Task	Hardware	Framework / precision	Seed	Checkpoint (SHA-256)
Parth	T1	RTX 5090 (31.8 GB)	torch 2.7.1+cu128 / bf16	8503	best.pt (ae9156f0...e859)
Chelsi	T1	RTX 3080 Ti Laptop (16 GB)	torch 2.11.0+cu128 / fp32	266	best.pt (d13501d8...095e)
Parth	T2	24-core CPU + RTX 5090	sklearn 1.9.1; torch 2.7.1 / fp32	8503	logreg_tfidf.joblib (5a9b2395...6a35); best_neural.pt (ef7925b9...eb99); best_neural_cnn.pt (7f34ff61...1614)
Chelsi	T2	RTX 3080 Ti Laptop	torch 2.11.0+cu128	266	baseline_mean.pt (7c7719a8...ecad); experimental_cnn.pt (9b8d57ab...a6ec); experimental_gru.pt (ae2ff39c...2c29)
Parth	T3	RTX 5090	torch 2.11.0+cu130 / FP32+TF32	8503	final_generators.pt (e979e900...5b92)
Chelsi	T3	RTX 4090	torch 2.6.0+cu124 / bf16	266	cyclegan_epoch_95.pt (a052a46b...c0f6; Git LFS)

Parth's raw logs and manifests are stored under the shared reproducibility directory, while Chelsi keeps the corresponding records within each task folder. Per-model metric reports contain the detailed confidence intervals, throughput, and memory measurements used in this report.

6. Overall Conclusions

Across the three tasks, architecture alone did not determine outcome: dataset choice, preprocessing, training budget, evaluation protocol, and checkpoint selection all affect how results should be interpreted. In Task 1, both character-level GPTs trained stably and showed the same greedy-versus-sampling trade-off, but different TinyStories variants prevent a controlled numerical ranking. In Task 2, within-member tests show that Parth's sparse TF-IDF baseline remained strongest on the full official test set, while Chelsi's BiGRU gave a statistically significant gain over her mean-pooling baseline. In Task 3, the members implemented nearly identical CycleGAN architectures with different training budgets and precision settings; Chelsi's run2 achieved the lower course-protocol FID and was submitted to Kaggle, while the internal metrics remain evaluator-specific.

The clearest lesson from the lab is that comparisons are most useful when the evaluation setup is shared. In a follow-up, we would standardize the test protocol, run multiple seeds, and keep a separate set for checkpoint selection before comparing models across members.

7. References

- [1] A. Vaswani et al., “Attention Is All You Need,” Advances in Neural Information Processing Systems 30, 2017.
- [2] R. Eldan and Y. Li, “TinyStories: How Small Can Language Models Be and Still Speak Coherent English?” arXiv:2305.07759, 2023.
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Appendix A. Additional Task 3 Metrics (Parth, EXP-05)

Table A1. Parth’s internal-evaluator and training-log metrics for EXP-05. These come from his own evaluator and are not comparable with Chelsi’s internal metrics.

Metric	Monet → photo (A2B)	Photo → Monet (B2A)
Precision / recall	0.5433 / 0.3558	0.3605 / 0.5367
Density / coverage	0.5778 / 0.0604	0.3268 / 0.9467
Cycle-reconstruction L1	0.0807	0.1018
LPIPS input → output / input → reconstruction	0.2987 / 0.2669	0.3801 / 0.2307
Content cosine (mean ± std)	0.8147 ± 0.0675	0.7525 ± 0.1058
Epoch-60 cycle / identity loss	0.0707 / 0.0689	0.0933 / 0.0854
Inference images/s	321.50	518.95

Training throughput was 28.266 images/s. At epoch 60 the mean gradient norms were 22.69 (G), 7.24 (D_A), and 6.98 (D_B), with no NaN steps.

Appendix B. Supplementary Figures

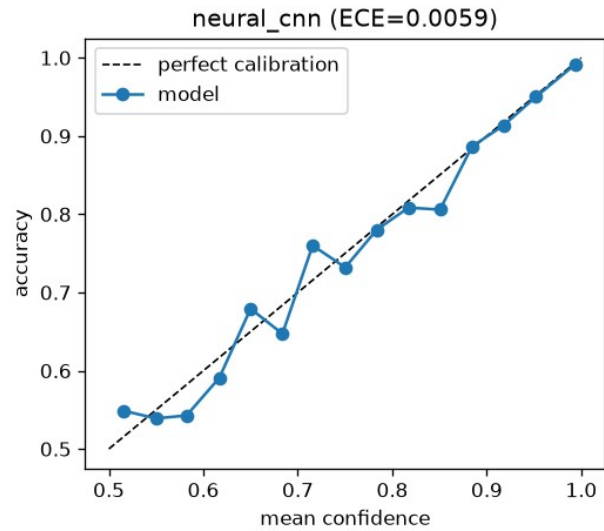


Figure B1. Reliability diagram for Parth's TextCNN (ECE 0.0059).



Figure B2. First six inputs per direction (sorted filenames) and the EXP-05 predictions scored by the course evaluator. The first four columns use the same source images as Chelsi's audit samples 1-4 (Figure 11).